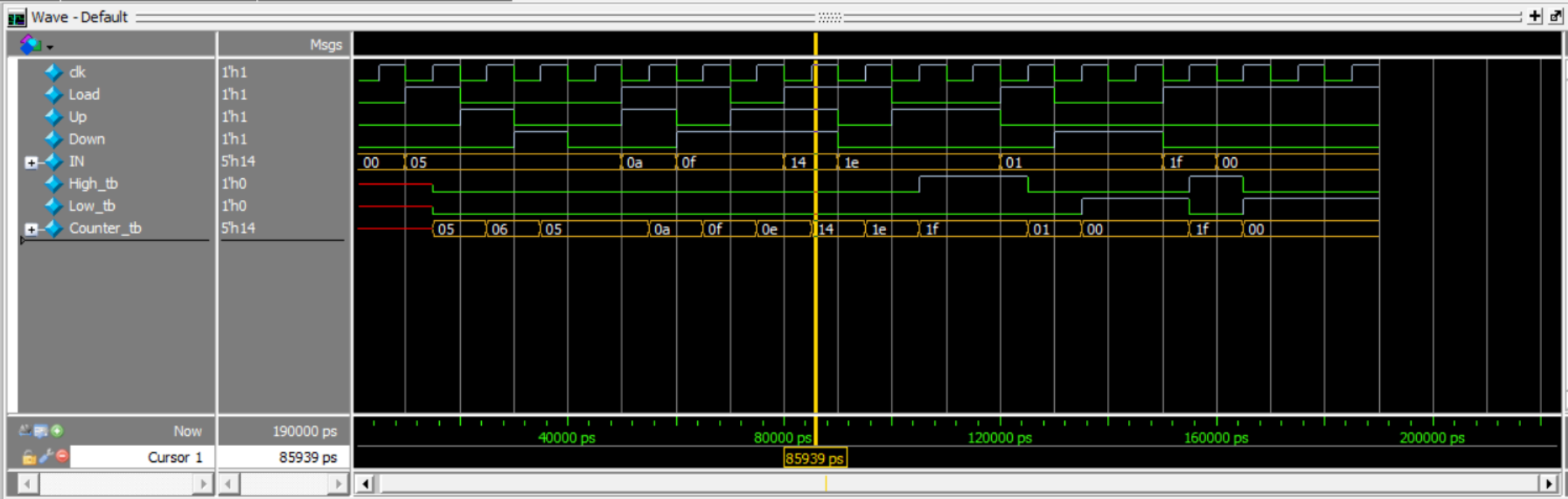
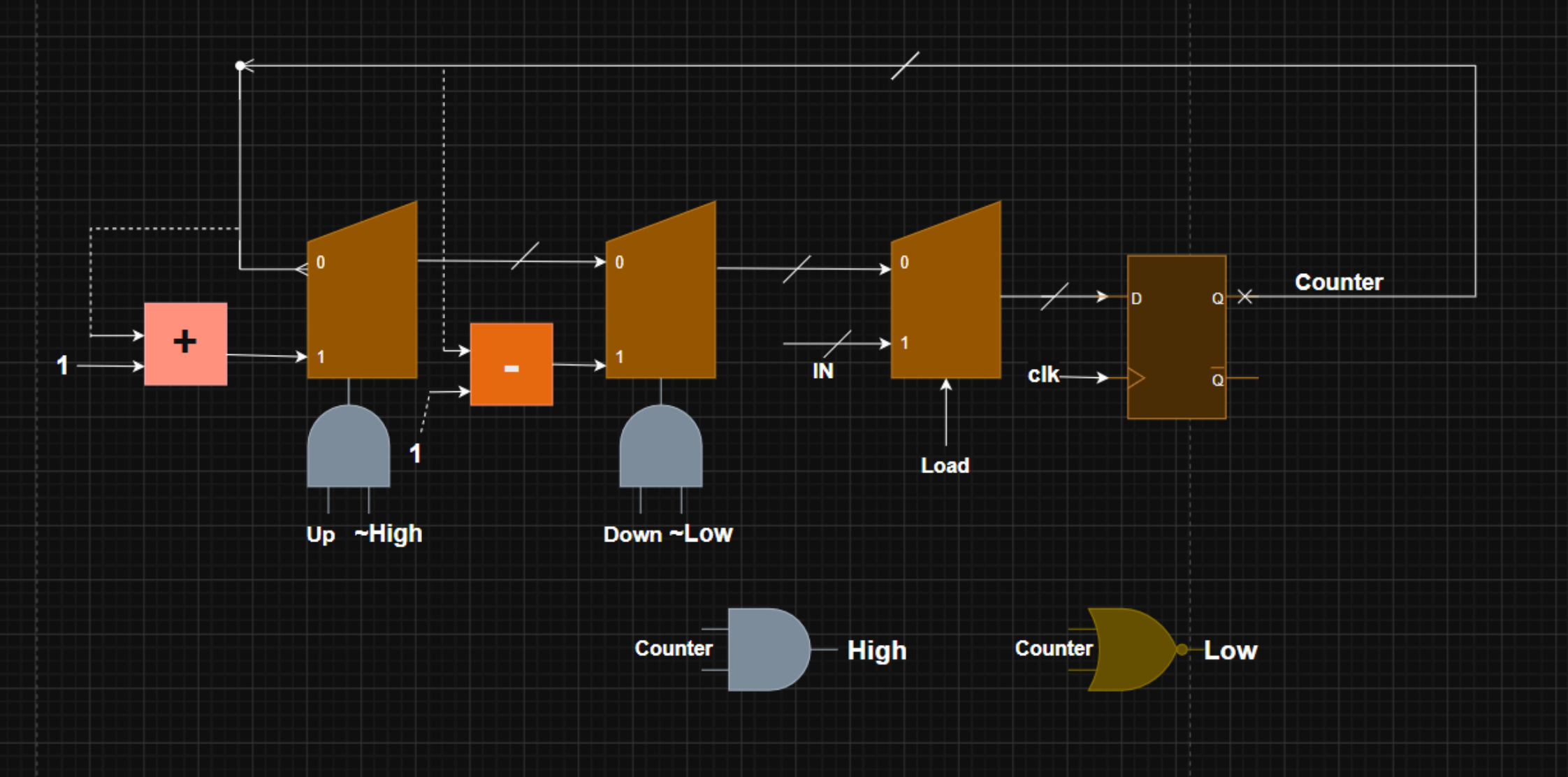
	A	B	C	D
1	Feature focus	Test Scenario	Inputs (Load, IN, Up, Down)	Expected Outputs (Counter, High, Low)
2	Initialization	Base Load	Load=1, IN=5, Up=0, Down=0	Counter=5, High=0, Low=0
3	Normal Count	Count Up by 1	Load=0, IN=X, Up=1, Down=0	Counter=6, High=0, Low=0
4	Normal Count	Count Down by 1	Load=0, IN=X, Up=0, Down=1	Counter=5, High=0, Low=0
5	Hold State	No control signals active	Load=0, IN=X, Up=0, Down=0	Counter=5, High=0, Low=0
6	Priority	Load overrides Up	Load=1, IN=10, Up=1, Down=0	Counter=10, High=0, Low=0
7	Priority	Load overrides Down	Load=1, IN=15, Up=0, Down=1	Counter=15, High=0, Low=0
8	Priority	Down overrides Up	Load=0, IN=X, Up=1, Down=1	Counter=14, High=0, Low=0
9	Priority	Load overrides ALL	Load=1, IN=20, Up=1, Down=1	Counter=20, High=0, Low=0
10	Upper Boundary	Normal count to Max	Load=0, IN=X, Up=1, Down=0 (Assume starting fr	Counter=31, High=1, Low=0
11	Saturation	Up at Max (No Rollover)	Load=0, IN=X, Up=1, Down=0 (Assume currently a	Counter=31, High=1, Low=0
12	Lower Boundary	Normal count to Min	Load=0, IN=X, Up=0, Down=1 (Assume starting fr	Counter=0, High=0, Low=1
13	Saturation	Down at Min (No Underflow)	Load=0, IN=X, Up=0, Down=1 (Assume currently a	Counter=0, High=0, Low=1
14	Flag Edge Case	Load Max directly	Load=1, IN=31, Up=0, Down=0	Counter=31, High=1, Low=0
15	Flag Edge Case	Load Min directly	Load=1, IN=0, Up=0, Down=0	Counter=0, High=0, Low=1
16				


Wave - Default
Msgs
clk
Load
Up
Down
IN
High_tb
Low_tb
Counter_tb
1'h1
1'h1
1'h1
1'h1
5'h14
1'h0
1'h0
5'h14
00
05
0a
0f
14
1e
01
1f
00
05
06
05
0a
0f
0e
14
1e
1f
01
00
1f
00
Now
190000 ps
Cursor 1
85939 ps
40000 ps
80000 ps
120000 ps
160000 ps
200000 ps
85939 ps

```
# [PASS]                                    Initialization (Load=5) | Counter: 5 | High: 0 | Low: 0 at time 20000
# [PASS]                                          Normal Count Up | Counter: 6 | High: 0 | Low: 0 at time 30000
# [PASS]                                        Normal Count Down | Counter: 5 | High: 0 | Low: 0 at time 40000
# [PASS]                                             Hold State | Counter: 5 | High: 0 | Low: 0 at time 50000
# [PASS]                              Priority: Load overrides Up | Counter: 10 | High: 0 | Low: 0 at time 60000
# [PASS]                            Priority: Load overrides Down | Counter: 15 | High: 0 | Low: 0 at time 70000
# [PASS]                              Priority: Down overrides Up | Counter: 14 | High: 0 | Low: 0 at time 80000
# [PASS]                             Priority: Load overrides ALL | Counter: 20 | High: 0 | Low: 0 at time 90000
# [PASS]                           Upper Boundary (Count to Max) | Counter: 31 | High: 1 | Low: 0 at time 110000
# [PASS]                                 Saturation (Up at Max) | Counter: 31 | High: 1 | Low: 0 at time 120000
# [PASS]                           Lower Boundary (Count to Min) | Counter: 0 | High: 0 | Low: 1 at time 140000
# [PASS]                               Saturation (Down at Min) | Counter: 0 | High: 0 | Low: 1 at time 150000
# [PASS]                        Flag Edge Case (Load Max directly) | Counter: 31 | High: 1 | Low: 0 at time 160000
# [PASS]                        Flag Edge Case (Load Min directly) | Counter: 0 | High: 0 | Low: 1 at time 170000
# ALL TESTS COMPLETED.
# ** Note: $finish    : Up_Dn_Counter_tb.v(130)
#    Time: 190 ns  Iteration: 0  Instance: /Up_Dn_Counter_tb
```


0
1
+
1
-
1
0
1
0
1
IN
Load
D
Q
Q
clk
Counter
Up
~High
Down
~Low
Counter
High
Counter
Low